

ACHIEVE!

SCIENCE



Ecosystems

Ally Chumley



Made for the
Australian Curriculum



Achieve! Science

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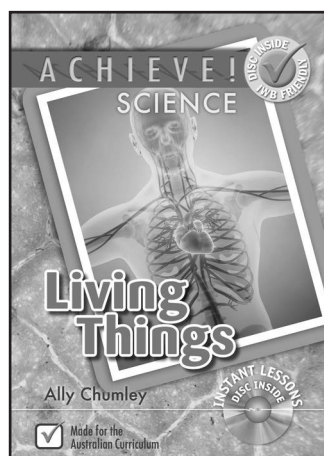
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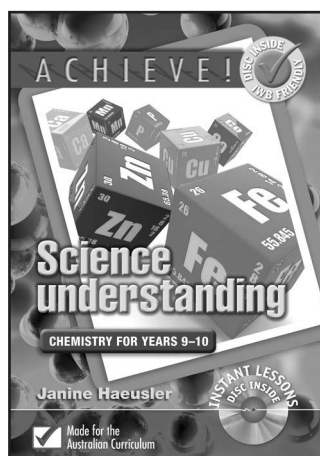
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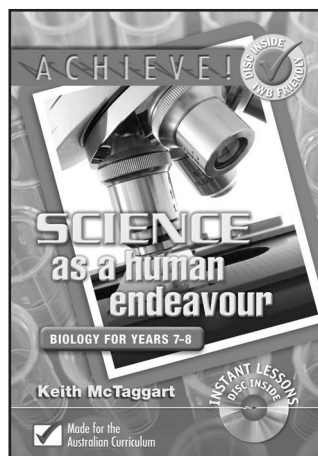
Living things

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A black and white photograph of a cheetah climbing a large, textured tree trunk. The cheetah is positioned in the center, with its body angled upwards and its head turned to the right. The tree trunk is thick and has a rough, cracked bark. The background is a plain, light gray.

A C H I E V E !

SCIENCE

Ecosystems

Ally Chumley

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Ally Chumley writes educational resources for teachers and learners across a range of subjects. She is an experienced teacher and enjoys the challenge of making difficult concepts accessible and inspiring in the classroom. Her formal credentials include a Bachelor degree and a Master of Education. Ally has written more than 40 titles for Australia's leading educational publishers. She lives in Comboyne, New South Wales, with her husband, two children and assorted farm animals.

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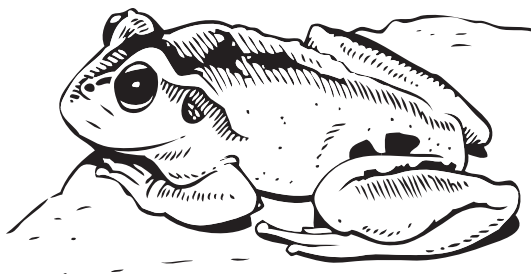
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The *Achieve!* series of science titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages will also be suitable as supplementary material for regular classes.

Ecosystems looks at how living things interact in ecosystems, food chains and food webs, habitats, animal and plant reproduction, adaptation, evolution by natural selection, and the effects of floods, drought and bushfires on Australian ecosystems.

This book is divided into six units, all of which contain photocopiable Activity sheets and a Revision sheet. The Activity sheets can be used sequentially or as standalone work sheets. Varied activities – including labelling diagrams, completing tables, multiple-choice questions, cloze activities and word games – allow students to consolidate their understanding and improve literacy skills. A background information box defines key terms and provides a pronunciation guide. Each unit concludes with a Revision sheet to test understanding of the unit content. Answers have been included for easy and efficient marking.

Many secondary school students who struggle with literacy have missed essential science concepts taught in primary school. *Ecosystems* addresses this issue by including core introductory concepts to help these students catch up.

The CD-ROM contains all the pages that are featured in the book. These pages can be easily edited to suit the individual needs of students. The CD-ROM also contains a folder of illustrations that can be used on other work sheets.

Australian Curriculum alignment		Unit					
	Year level	1	2	3	4	5	6
Living things, including plants and animals, depend on each other and the environment to survive (ACSSU073)	4	✓	✓	✓	✓		
Living things, including plants and animals, depend on each other and the environment to survive (ACSSU073)	5	✓					
The growth and survival of living things are affected by the physical conditions of their environment (ACSSU094)	6	✓					
Interactions between organisms can be described in terms of food chains and food webs; human activity can affect these interactions (ACSSU112)	7		✓				✓
Multi-cellular organisms contain systems of organs that carry out specialised functions that enable them to survive and reproduce (ACSSU150)	8					✓	✓
Ecosystems consist of communities of interdependent organisms and abiotic components of the environment; matter and energy flow through these systems (ACSSU176)	9	✓	✓	✓	✓		
Chemical reactions, including combustion and the reactions of acids, are important in both non-living and living systems and involve energy transfer (ACSSU179)	9			✓			
The transmission of heritable characteristics from one generation to the next involves DNA and genes (ACSSU184)	10					✓	
The theory of evolution by natural selection explains the diversity of living things and is supported by a range of scientific evidence (ACSSU185)	10					✓	

What is an ecosystem?

An ecosystem is a complex **system** that includes all the plants, animals and non-living things found in a certain place, as well as all the different ways those things **interact**. We say that something interacts with something else when it acts in a way that affects the other thing.

For example, a garden pond contains living things – also called **organisms** – such as frogs, snails, fish, water birds, water plants and algae, as well as non-living things such as fresh water, rocks, sand, air, sunlight and soil. The garden pond ecosystem is all of these living and non-living things as well as all of the ways these things interact and depend on each other – from a fish eating a snail to water lilies using sunlight to make food. All the living things in an ecosystem are known as a **community**. The variety of living things in the ecosystem is known as the ecosystem’s **biodiversity**.

 **1** Draw a line between each word and its meaning.

Word	Meaning
a) ecosystem	the variety of different lifeforms in a particular place or ecosystem
b) system	all the living things that live in the same place and depend on one another
c) interact	act in ways that affect one another
d) organism	the living and non-living things in a certain place and all their interactions
e) community	a set of connected things that form a whole
f) biodiversity	a living thing

2 Think of the ways that animals, plants and non-living things interact in a garden pond ecosystem. List four interactions in the answer spaces below, and then share your answers with a partner. One has been done for you.

A frog eats an insect.

What is an ecosystem? CONT'D

Balanced ecosystems

An ecosystem is described as balanced, or **healthy**, when all the groups of living things are **stable** and nothing is disturbing the flow of energy and nutrients. Human activities – such as cutting down trees, introducing new animal species and planting crops that are not native to an area – can affect the delicate balance of an ecosystem. Other things that may affect ecosystems are floods, fires, drought and changes in climate.

3 Fill in the table by giving examples of how these things might disturb the balance in an ecosystem. One has been done for you.

Disturbance	Possible effect
cutting down trees	destroys the homes of some animals, causing them to move away or die out
introducing new plant or animal species	
flood	
drought	

More about ecosystems

-  **1** Unscramble the letters in brackets, and then write in the new words to complete this general information about ecosystems.

Living creatures of all kinds need (doof) _____, water and shelter. To obtain food, some living things (tea) _____ other living things in their ecosystem. We show these relationships in a (diamarg) _____ called a food web.

In every ecosystem, there is a source of drinking (tawer) _____. It may be a (ereck) _____ or pond or it may be deep under the (drongu) _____.

Most living things need clean (ria) _____ to breathe, as well as the right amount of (tighl) _____ or darkness and a way of keeping their body temperature warm or (loco) _____ enough. To shelter themselves and their offspring, animals make (shome) _____ in the plants, rocks, soil and other natural features of an ecosystem.

An animal's ideal home is often a place that offers protection from predators, shelter from harsh weather conditions and a feeling of (tafesy) _____.

The environment most suitable for a particular living thing's survival is called its (bitahat) _____.

Energy and nutrients

All the living things and non-living things in an ecosystem are involved in the flow of **energy** and **nutrients**. Energy enters the ecosystem as sunlight (light energy) and is made into food by green plants. Some animals eat the plants and other animals eat these animals. When animals and plants die, organisms such as fungi break down the remains into small parts called nutrients, which go into the soil and are then used by new green plants. The **cycle** begins again.

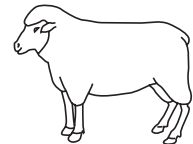
- 2** Draw a line between each sentence starter and the most appropriate sentence ending.
- | | |
|-------------------------------------|---|
| a) Green plants | eat plants and release the energy in the food. |
| b) Animals | use energy from sunlight to create their own food. |
| c) Decomposers such as fungi | break down animal and plant remains into small parts. |

More about ecosystems CONT'D

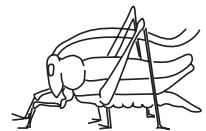
- 3 a) Circle four or more pictures that show living things that might come from the same ecosystem. Name the ecosystem.



- b) Use a different colour to circle another four or more pictures that show living things from another ecosystem. Name the ecosystem.



- c) Which, if any, of the living things are included in both your ecosystems?



- 4 Write at least two plants found in each type of ecosystem listed below. The first one has been done for you.



a) a snowy, alpine forest fir trees, moss

b) a hot, dry desert

c) the Australian bush

d) a garden pond

e) a tropical rainforest

What is a habitat?

BACKGROUND

species (say ‘spee-seas’) – a group of living things that can breed together to produce fertile offspring

The particular place where a **population** of a living thing lives is called its **habitat**. A population is a group of animals from the same species that lives in the same space, such as a population of gorillas in a mountain range. A habitat can be as big as a rainforest or as small as a burrow, depending on how much space the population needs.

Many living things need to be able to find food, water, clean air, shelter, warmth, light and safety in their environment. When one or more of these basic needs is no longer available – for example, if drinking water is no longer available in drier months – the population may sometimes move to a new habitat.

 **1** How do each of these living things meet some of their needs? Complete the table by writing these phrases into the correct spaces.

fresh grass to graze on eucalyptus leaves fruit trees

sharp claws and teeth coral reef a protective family group

underground burrows protection of a sea anemone

Living thing	Food	Shelter	Safety from predators
rabbit			burrows with hidden entrances
clownfish	tiny fish, plankton		
spider monkey		leafy beds in the high rainforest canopy	
koala		tall trees high off the ground	

What is a habitat? CONT'D

2 Use the code to find the things that most living things need from their environment.

Code												
a	b	c	d	e	f	g	h	i	j	k	l	m
8	1	15	11	17	26	2	10	19	20	3	12	21
n	o	p	q	r	s	t	u	v	w	x	y	z
4	25	13	16	9	7	22	18	6	24	23	14	5

- a) — — — — — — —

7 10 17 12 22 17 9
- b) — — — —

26 25 25 11
- c) — — — — — — — —

15 12 17 8 4 8 19 9
- d) — — — — —

24 8 22 17 9

3 Use the information in the table in Question 1 to describe the habitat for each of the following animals. Use your general knowledge to add more information.

- a) rabbit habitat
-
- b) clownfish habitat
-
- c) koala habitat
-

Ecosystems and habitats

 1 Fill in the missing vowels to complete the table.

Key term	Definition
__ c __ syst __ m	the living and non-living things in a certain place and all their interactions
h __ b __ t __ t	the particular place where a population of a living thing lives
c __ mm __ n __ ty	a group of living things that live in the same place and depend on one another
p __ p __ l __ t __ __ n	a group of animals from the same species that lives in the same space

- 2 Underline the correct word within each set of brackets to complete the sentences.
- a) An (ecosystem / organism) may be made up of one (habitat / living thing) or many.
 - b) Within an (ecosystem / habitat), different populations of living things interact and form a (population / community).
- 3 a) Give two examples of an ecosystem. (For example, a rainforest ecosystem.)
-
-
- b) Give two examples of a type of habitat. (For example, a tree.)
-
-
- 4 a) Circle one ecosystem in your answers for Question 3a. Name two living things and two non-living things that are part of this ecosystem.
-
-
-
-
- b) Circle one habitat in your answers for Question 3b. Name two species populations that live in this type of habitat.
-
-

Changing behaviour

BACKGROUND

habitat (say 'hab-i-tat') – the particular place where a group of living things live

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

Adaptation is the name that scientists give to the way that living things change and become better suited to their habitats. They also use the word adaptation to describe the actual change.

Adaptations may be behavioural or physical. Physical adaptations in species happen over many generations, but a species can change its behaviour more easily. This is called **behavioural adaptation**.

 **1** What word do we use to describe the way that some organisms go through changes that help them become better suited to their environment?

2 What smaller (root) word can be found in this term? a ____ p ____

3 What does the word in Question 2 mean? Circle one answer:

- a)** to change
- b)** to die
- c)** to stay the same

Many organisms can make outward changes to suit their environment. For example, long-haired dogs such as Siberian huskies are able to keep warm in cold habitats because of their thick coats. When a dog like this is placed in a warmer habitat it must change its behaviour so it can cope with the different temperature. It changes its behaviour, so this is behavioural adaptation.

4 Circle three behaviours that Siberian huskies could develop to help them stay cool in a warm habitat.

- | | |
|--|------------------------------|
| grow finer, shorter hair | drink more water |
| develop scales | lie in the shade |
| limit movement at certain times of the day | increase its amount of blood |

Relationships in ecosystems

There are many types of relationships between living things in an ecosystem. Some relationships place two living things against each other (for example, the relationship between an animal that hunts and eats another animal). Other relationships can **benefit** one or both organisms.

Mutualism is a type of relationship between different living things that benefits both. For example, a small fish that eats a special type of algae lives amongst a group of bigger fish that have this algae growing on their skin. The smaller fish cleans up the skin of the bigger fish by nibbling at the algae. Both types of fish benefit. The little fish gets fed, and the bigger fish gets rid of its skin problem.

 **1** What is mutualism? Use the words in the box to write a definition.

a relationship	living things	benefit	between	where
organisms	describes	mutualism	both	

2 Are the following statements true or false?

- | | | |
|--|------|-------|
| a) Mutualism does not benefit small organisms. | True | False |
| b) Relationships always benefit one organism and not the other. | True | False |
| c) Living things can sometimes live together in a way that helps both things. | True | False |
| d) Mutualism only exists amongst certain fish populations. | True | False |

Relationships in ecosystems CONT'D

- 3 Complete this passage about mutualism by inserting the words from the word bank.

Word bank

skin
covered
food
algae
mutualism
benefit
smaller

A certain fish that has algae growing on its _____ lives with a group of _____ fish that eat the _____ off the skin. This relationship enables the larger fish to keep its skin from being completely _____ by the algae. The relationship between these fish is an example of _____, where both living things _____ from the relationship: the small fish get _____ and the larger fish gets regularly cleaned.

- 4 Read Question 3 again, and then unscramble the words in brackets to complete this passage.

There is a third (ganisorm) _____ mentioned in this example: the (galea) _____ that lives on the (knis) _____ of the larger fish. This is an example of another type of (shiplaretion) _____, called a parasitic relationship.

- 5 Are these relationships an example of mutualism? Circle yes or no.

a) A lybia crab, or pom-pom crab, lets sea anemones grow on its claws. The anemone gets carried around and can find food more easily. When a predator approaches, the crab waves the stinging sea anemones like pom-poms to defend itself.

Yes No

b) Barnacles attach to the underside of a baleen whale. The barnacles feed well on the many nutrients that are available there. The whale is unaffected by the barnacles.

Yes No

Competition in ecosystems

Ecosystems do not always operate in peace and harmony. The living things in an ecosystem have different needs, which cause them to **compete** in a number of ways. Animals compete for food, water and shelter in order to survive. They also compete amongst members of their own species for mating partners and they fight to protect their offspring.

Even plants compete in order to survive. Plants produce their own food by using sunlight and other ingredients to make sugars. In an area crowded with plants, they must compete with the other plants for sunlight – for example, rainforest plants grow taller and taller so they can rise above the rainforest canopy (roof of leaves) and reach the sunlight. Plants must compete for growing space, water and nutrients in the soil. They also compete to attract insects that help them to reproduce.

 **1** Correct the spelling of these phrases by rewriting them in the spaces provided.

- a) piece and harmany _____
- b) in awder to sirvive _____
- c) hellp them to reeproduce _____

2 Circle the two statements that mean the same thing.

- a) Plant and animal species must compete in order to survive.
- b) Plants and animals live to compete against each other.
- c) Different types of plants and animals need to fight for things so that they can live.

3 a) List four things that animals compete for.

_____	_____
_____	_____

b) List four things that plants compete for.

_____	_____
_____	_____

Competition in ecosystems CONT'D

4 Unscramble each of the two sets of mixed-up sentence parts and then rewrite each sentence in the correct order.

a) are Ecosystems + always not peaceful

_____.

b) different ways compete in + things Living

_____.

c) compete Animals + mating partners for

_____.

d) for sunlight + with each other Plants compete

_____.

e) growing space Plants need + and soil from the nutrients

_____.

5 Look at the sentences you have written in Question 4. Write an example that illustrates each fact. The first one has been done for you.

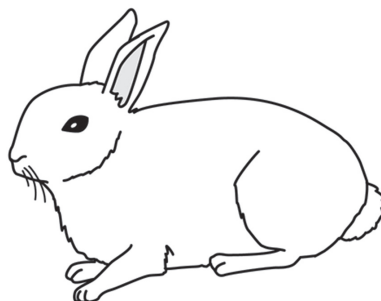
a) For example, rabbits are hunted by foxes in woodland ecosystems.

b) For example, _____.

c) For example, _____.

d) For example, _____.

e) For example, _____.



Check your understanding

Circle the correct answers to these revision questions.

- 1 Another name for a living thing is
 - a) an ecosystem.
 - b) a habitat.
 - c) an organism.
- 2 One example of an ecosystem is
 - a) a rainforest.
 - b) the animal kingdom.
 - c) reptiles.
- 3 Which of the following is not a basic need of most living things?
 - a) water
 - b) food
 - c) money
- 4 What do some animals do when they cannot find shelter, food or water?
 - a) They move to Antarctica.
 - b) They move to a new habitat that is more suitable.
 - c) They change themselves so they have different needs.
- 5 What is an example of behavioural adaptation?
 - a) dying your hair
 - b) drinking more water in a hotter climate
 - c) breeding new characteristics
- 6 Which of the following is an example of mutualism in action?
 - a) when a small fish is eaten by a larger fish
 - b) when a fish eats the algae off another fish, and both fish benefit
 - c) when a large fish kills and eats a smaller fish
- 7 How do animals compete with other animals in reproduction?
 - a) They compete to attract the best mating partner and to defend their offspring.
 - b) They compete for sunlight.
 - c) They compete for plants to feed their offspring.
- 8 Which plants win the competition for sunlight?
 - a) the plants that are the closest to the forest floor
 - b) the middle-sized plants
 - c) the plants whose leaves form the canopy of the forest

Producers, consumers and decomposers

All the organisms in an ecosystem can be grouped into one of three types, depending on what they eat.

- **Producers** are living things that make (or produce) their own food from inorganic (non-living) ingredients. Examples of producers are green plants.
- **Consumers** are organisms that cannot make their own food. They either eat or use food products made by producers or other living things that eat producers. Examples of consumers are insects and snakes.
- **Decomposers** break down organic matter (dead plants and animals) by making it rot or decay, releasing nutrients, and then absorb the nutrients as food. Examples of decomposers are bacteria and fungi. Many decomposers are microscopic organisms, called microorganisms.

- 1 a) In the diagram below, colour any producers GREEN and any consumers RED.
- b) Can you see any decomposers? Why or why not?



Producers, consumers and decomposers CONT'D

2 Complete the table by giving examples of producers, consumers and decomposers for each ecosystem. Some examples have been given. Work alone or with a partner.

	Australian desert ecosystem	African savannah ecosystem
Producers	spinifex grass kangaroo paw Sturt's desert rose	
Consumers		
Decomposers		fungi bacteria termites

3 Write definitions for the following words. Use your dictionary if needed.

- a) organic _____
- _____
- b) inorganic _____
- _____

What is a producer?

BACKGROUND

consumers (say 'con-syoo-mers') – living things that eat or use food products made by producers or that eat other consumers

decomposers (say 'de-com-poz-ers') – living things that work to break down once-living materials into smaller parts

Producers are living things that make, or produce, their own food from inorganic (non-living) ingredients. Examples of producers are certain grass plants, vines and trees. Some of the inorganic ingredients that they use to make food include carbon dioxide and water.

Consumers and decomposers depend upon the food that producers make to provide them with energy.

 **1** Use full sentences to answer these questions about producers.

a) Are humans producers or consumers?

b) Producers use what type of ingredients to produce their food?

c) What does the word *produce* mean?

d) What does the word *inorganic* mean?

2 a) List four species of animals that rely on plants that produce fruit.

b) List four species of animals that need to eat grass in order to survive.

What is a producer? CONT'D

3 Which of these foods that humans eat are produced by plants? Circle your answers.

wheat	cherries	watermelon
butter	corn	bacon
beans	milk	tomatoes
ice cream	eggs	onions
potatoes	jelly	bananas
apples	steak	carrots
grapes	sesame seeds	olives
chicken	oats	fish
rice	turkey	celery

4 Which producer plants do each of these foods and drinks come from?

- a) sugar _____
- b) peanut butter _____
- c) coffee _____
- d) tea _____
- e) orange juice _____
- f) soy milk _____
- g) cornflakes _____
- h) olive oil _____
- i) wine _____
- j) pasta _____

5 What do you think would happen if half of the producers in an ecosystem died out? How would this affect consumers? Give an example.

What is a consumer?

BACKGROUND

mammals (say 'ma-mels') – warm-blooded animals with backbones that feed their young with milk

producers (say 'pro-dews-ers') – living things that make their own food from non-living ingredients

Consumers are types of organisms that eat or use food products made by producers or that eat other consumers. Large land mammals, birds and reptiles are consumers.

There are three main types of consumers:

- **herbivores** (consumers that eat plants)
- **carnivores** (consumers that eat animals, or meat)
- **omnivores** (consumers that eat both plants and animals).

Humans are omnivores. We eat all sorts of foods produced by plants, and we also eat other animals that eat plants and animals.

 **1** List six types of living things that are consumers.

_____	_____
_____	_____
_____	_____

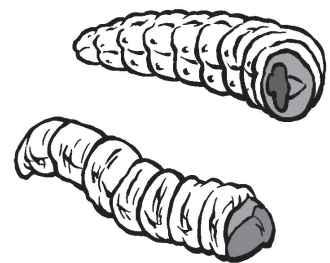
2 Decide whether these things are producers or consumers. Circle your response for each one.



a producer OR consumer



b producer OR consumer



c producer OR consumer



d producer OR consumer



e producer OR consumer

What is a consumer? CONT'D

Most carnivores are **predators**. Predators hunt, kill and eat other living things. The living thing that a predator hunts is called its **prey**. For example, the African lion is a fierce predator that needs meat to survive. Its prey includes wildebeest, zebras and giraffes.

- 3 a) Which of the living things in the box are predators? Underline your choices.
- b) Which of the living things in the box are prey for different types of predators? Circle your choices.

tiger	caterpillar	snail	zebra
brown bear	mouse	rabbit	dingo

- c) Look at the living things you have underlined. List each one below and write whether that predator is a carnivore or omnivore.

Predator: _____ Carnivore or omnivore? _____

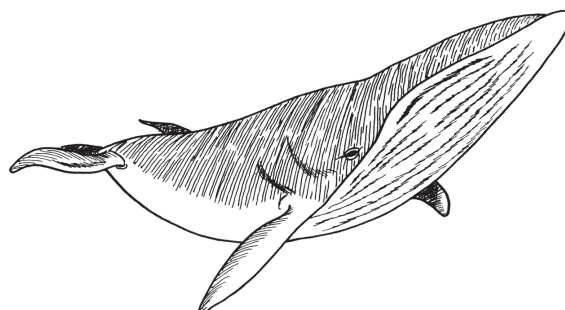
Predator: _____ Carnivore or omnivore? _____

Predator: _____ Carnivore or omnivore? _____

- 4 a) Which of these animals have no natural predators to fear in their ecosystems? Circle your choices.

blue whale	lizard	prawn
rabbit	crocodile	lion

- b) Look at the animals you have circled. What animal hunts and kills these animals, but is not considered a natural predator?



What is a decomposer?

BACKGROUND

nutrients (say 'nu-tree-ants') – substances that give animals and plants the energy essential for growth and life

A decomposer is a living thing that works to break down **organic** materials into smaller parts. For example, a decomposer breaks down fruit on a forest floor by making it rot or decay. The decomposer absorbs the nutrients from the remains and also releases some nutrients into the soil.

Fungi and moulds are decomposers. They decompose plant waste, including vegetables. Some small organisms from the animal kingdom are also decomposers, including worms, slugs, millipedes and termites, which break down logs and sticks. Other decomposers, like maggots and flies, help to break down dead animal carcasses (bodies). Dung (animal droppings) is broken down by dung flies and beetles. Microorganisms, including various bacteria, break down all kinds of substances, including milk, eggs and meat products.

 **1** Circle six things that are classified as decomposers.

termite	bacteria	mould	toadstool
maggot	rabbit	worm	eagle

2 Write these words into the spaces below. The lines break them into syllables.

decompose _ _ _ / _ _ _ _ / _ _ _ _ _

decomposers _ _ _ / _ _ _ _ / _ _ _ _ / _ _ _ _

decomposition _ _ _ / _ _ _ _ / _ _ _ _ / _ / _ _ _ _ _

3 Complete this passage by filling in the missing vowels (a, e, i, o, u).

D _ c _ mp _ s _ rs break down dead _ rg _ n _ sms into sm _ ller parts.

Leaf litter, animal d _ ng, bodies of d _ _ d creatures and dead pl _ nts all

d _ c _ mp _ s _ . After decomposition, the n _ trients that were contained in the decayed organisms return to the s _ _ l. In this way, they are r _ cycl _ d. Nutrients in the soil are then taken up into y _ _ ng plants and _ n _ mals eat the plants. The cycle of life and death begins again.

What is a decomposer? CONT'D

4 Match the decomposer with the material it works on by drawing a line between the two.

Material	Decomposer
a) leaf litter	dung flies
b) logs	slugs
c) dung	lactobacteria
d) fruit	maggots
e) dead animal carcass	termites
f) vegetables	fungi
g) milk	fruit flies

In an ecosystem, the substances contained in food are cycled through the system by producers, consumers and decomposers – although the substances change their form. The main chemical substance in food and living things is called **carbon**.

5 Unscramble the words in brackets to complete the passage.

Dead leaves, grass and (ruitf) _____ and (egetablesv) _____ begin to (ecayd) _____ and become slimy and (mellys) _____ on the ground. (ecomposersD) _____ act on these items in the (armw) _____, (arkd) _____ soil. These decomposers include (acteriab) _____, (ouldm) _____ and (ungif) _____. They (reakb) _____ down the material and the (utrientsn) _____ return to the soil. Nutrients are made of (hemicalsc) _____ that living things need to eat and (bsorba) _____ to stay healthy. (anureM) _____ and (ottenr) _____ (eatm) _____ from the (arcassesc) _____ of dead animals are another source of food for decomposers. They break these down and the (arbonc) _____ cycle continues in the (cosysteme) _____.

Food chains

BACKGROUND

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

In an ecosystem, one organism depends on another to survive. Producers make food, which is then eaten by consumers, which is then broken down by decomposers. Consumers and decomposers turn this food into the energy they need to survive. This creates what scientists call a **food chain**.

A food chain can be represented using pictures or words, but it always needs arrows. The arrows show what is eaten by what and the **flow of energy** along the food chain. They point away from the thing that is being eaten. For example:

plant → grasshopper → snake → eagle

Sometimes food chains show decomposers. Sometimes they show just producers and consumers, or just consumers.

 **1** Read these statements and then write a food chain for each. Two examples have been done for you. Hint: Make sure your organisms are in the correct order. The correct order may not be the order in which they appear in the statement. Check your answers by replacing each arrow in the food chain with 'is eaten by'.

- a)** A scorpion is eaten by a meerkat, which is eaten by a cobra, which is eaten by a vulture.

scorpion → meerkat → cobra → vulture

- b)** A tiger eats a deer, which eats a plant.

plant → deer → tiger

- c)** A ferret eats a rabbit, which eats grass.

- d)** A brown bear eats a salmon, which eats plankton.

Food chains CONT'D

- e) Seaweed is eaten by a small fish, which is eaten by a tuna, which is eaten by a cormorant.

- f) A lynx eats a raccoon, which eats a rat, which eats an insect.

- g) A wolf eats a woodpecker, which eats a frog, which eats a fly.

- h) Grain is eaten by a fieldmouse, which is eaten by a weasel, which is eaten by a cougar.

- i) A shark eats a jewfish, which eats a whiting, which eats a prawn, which eats a worm.

- j) Fruit is eaten by a grub, which is eaten by a monkey, which is eaten by a hyena.

- k) A crab eats a sea snail, which eats a starfish, which eats a mussel, which eats plankton.

Food chains are not the best way to describe the relationships in most ecosystems. They are too simple. Ecosystems usually involve many species that are connected in all sorts of ways.

For example, this food chain is from an African savannah: grass → zebra → lion. There are many savannah animals that consume grass, and many that are eaten by lions. Birds also eat the remains of zebras, and decomposers like bacteria, flies and microbes eat them as well. The best way to show all of these relationships is using a **food web**.

Food webs

BACKGROUND

habitat (say 'hab-i-tat') – the particular place where a group of living things live

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

Instead of drawing or writing a food chain, we can better show the many relationships between living things in an ecosystem by drawing a food web.

There is often more than one species that eats another in an ecosystem. Food webs are used to describe the diets of animals that have a variety of food sources, and they are also used to show all the relationships between creatures in a particular ecosystem.

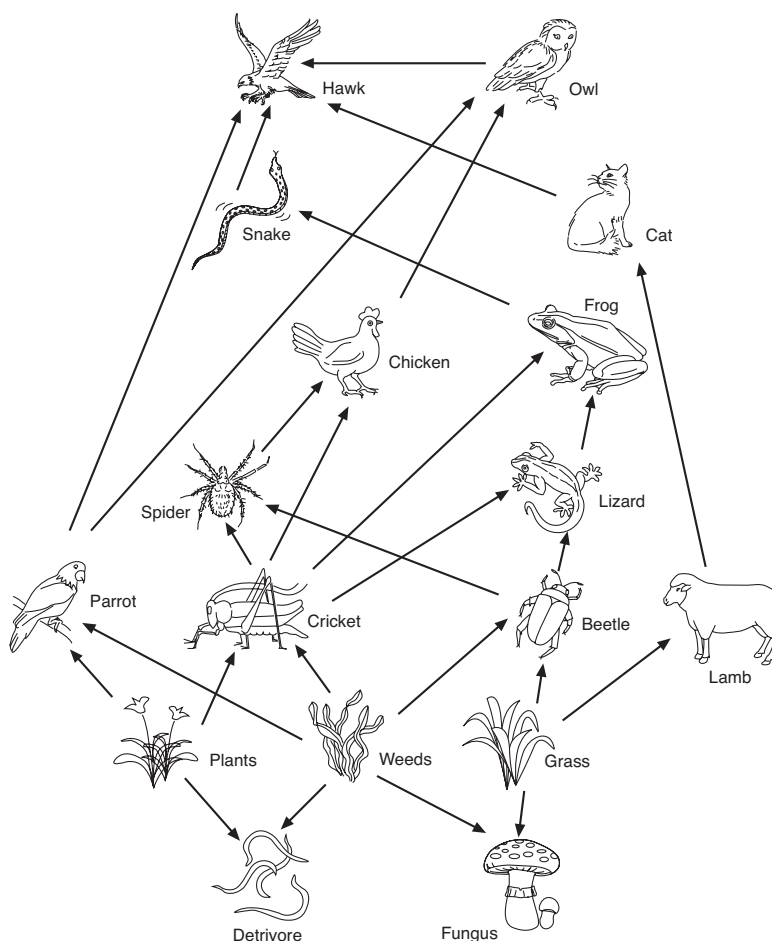
 **1** Look at the food web pictured and answer the following questions.

a) What eats the frog?

b) What is eaten by the chicken?

c) What is consumed by the cricket?

d) What kind of ecosystem do you think this food web comes from?



Food webs CONT'D

- 2 a)** Complete the passage below by writing in the missing words. The words you need are in the letter chain in the box. Separate the words, then write each one into the spaces in order.

foodsourceskillingsmallereggplantsillnessattack

Some creatures may get their _____ from a variety of _____. This does not always involve _____ prey. For example, a wild dog may kill a _____ animal as one way of getting food. But it may also hunt for _____ in nests, eat _____ or even eat the bodies of animals that died from _____ or from an _____ by another animal.

- b)** In your workbook, draw a small food web that shows everything that a wild dog eats. Connect each source of food to the wild dog, using arrows.
- 3** In your workbook, plan and draw a food web for a coral reef. Include the organisms listed in the box below.

Steps to take:

- Make a list of any other living things you wish to include.
- Draw or write food chains to show what eats what.
- Check if there is more than one predator for each type of prey.
- Sketch a plan for your food web before drawing your final version.

plankton	small fish	large fish
sharks	whales	turtles
seaweed	sea slugs	crabs
prawns	sea birds	worms

Check your understanding

Circle the correct answers to these revision questions.

- 1 The three types of organisms in most habitats are
 - a) producers, directors and actors.
 - b) producers, customers and composers.
 - c) producers, consumers and decomposers.
- 2 Which of the following is an example of a consumer in action?
 - a) a tree growing fruit
 - b) a goat eating grass
 - c) a cornstalk growing in the garden
- 3 What is the main chemical substance that makes up food and living things?
 - a) carbon
 - b) oxygen
 - c) dung
- 4 What type of living things can break down milk and eggs, making them go rotten and decay?
 - a) vegetables
 - b) bacteria
 - c) cows
- 5 A predator is a type of
 - a) producer.
 - b) decomposer.
 - c) consumer.
- 6 Which of these is the most complex way in which we can show food relationships in ecosystems?
 - a) food webs
 - b) food chains
 - c) food stores
- 7 Which of these food chains correctly shows that a zebra, which eats grass, is eaten by a lion?
 - a) grass → zebra → lion
 - b) grass < zebra < lion
 - c) lion → grass → zebra
- 8 What is the difference between a food chain and a food web?
 - a) a food chain is longer, showing many different types of predators and their prey
 - b) a food chain shows only decomposers
 - c) a food chain is simpler, showing only one main source of food for each living thing

The importance of plants in ecosystems

Plants are a very important part of ecosystems for a number of reasons. One reason is that they are producers, which means they make their own food from inorganic (non-living) ingredients, and other living things consume this food. Another important reason is that as they are making food they take in **carbon dioxide** and release **oxygen**.

Green plants take in carbon dioxide gas through their leaves. Carbon dioxide, water and sunlight are used during a process called **photosynthesis** to help the plant produce a sugary food called glucose. When the plant breaks down water during photosynthesis, one of the waste products is oxygen. The plant releases the oxygen through its leaves. Most animals, including humans, need oxygen in the air they breathe for **respiration**.

The word equation for photosynthesis is:

carbon dioxide + water (+ light energy) = glucose + oxygen

The chemical formula for carbon dioxide gas is CO_2 (one part carbon and two parts oxygen).

The chemical formula for water is H_2O (two parts hydrogen and one part oxygen).

 **1** Write the answers to these questions in the spaces provided.

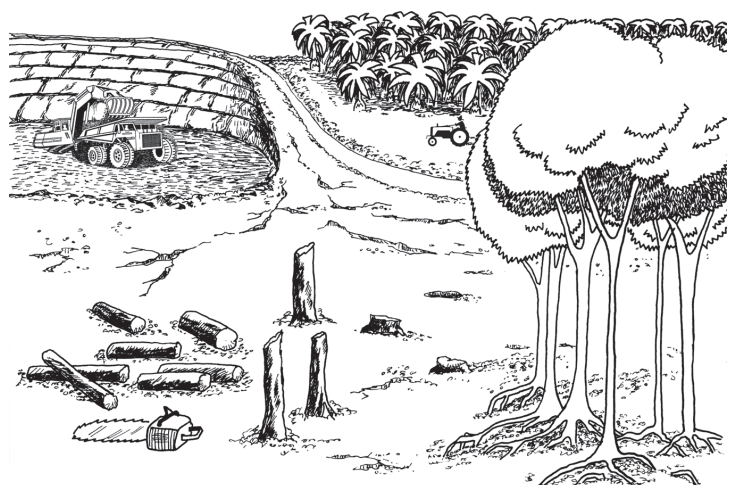
- a) What is the name of the gas that plants take in? _____
- b) Which parts of the plant collect the gas? _____
- c) In what process is the gas used by the plant? _____
- d) What two substances result from this process? _____
- e) One of these substances is a waste product, not used by the plant. Which substance is this? _____
- f) Why do most animals need oxygen? _____
- g) Which chemical element is represented by C? _____
- h) Which chemical compound is CO_2 ? _____

The importance of plants in ecosystems CONT'D

- i) What common liquid is H_2O ? _____
- j) Which two elements make up water molecules? _____
- 2 What would be some of the effects on a forest ecosystem if more than half the trees and plants were cut down or destroyed? List five effects, and underline who or what would be affected. Share your answers with a partner. One has been done for you.
- Birds such as woodpeckers would lose their nests or homes.
- _____
 - _____
 - _____
 - _____
 - _____
- 3 According to this Activity sheet, what are the two main reasons why plants are important for ecosystems?
- _____
- _____



Before logging



After logging

Photosynthesis and ecosystems

Green plants produce their own food using photosynthesis. This food helps them grow. As the plant grows, more leaves, fruit and flowers are made available for other living things to eat. In this way, photosynthesis helps an ecosystem to thrive.

Plants use carbon dioxide, water and the sun's energy to make a sugary food called glucose. When the plant is eaten by a consumer, some of the energy stored in the glucose is released and used by the consumer. The rest of the energy is released as heat. When consumers and producers die, decomposers break down their remains into tiny parts called nutrients. These nutrients go into the soil or the atmosphere. A new plant uses these nutrients to begin the **energy transfer process** all over again.

-  **1** Write the words in the word bank in alphabetical order.

Word bank

photosynthesis
ecosystem
carbon dioxide
water
energy
glucose
nutrients

- 2 a)** There are two parts to the word *photosynthesis*. *Photo* means 'light' and *synthesis* means 'to make'. What do plants use light to make?

- b)** Other English words use the prefix (word beginning) *photo*. List at least three.

- 3** In your own words, describe the function of photosynthesis in ecosystems.

Photosynthesis and ecosystems CONT'D

3 Photosynthesis occurs in green plants. Cross out the letters p h o t o s y n t h e s i s , in order, to find the names of the plant parts that do each job. Write the plant parts in the table. The first one has been done for you.

Jobs	Plant part
collect carbon dioxide	phloetaovsyntheesiss leaves
attract insects for pollination (the process that allows some plants to reproduce)	fphlotosowyntheseriss _____
store seeds	pfhotorusynthesist _____
provide support and strength	phsotostynthaesislk _____
collect water	phrootoosynthesists _____

4 Label the diagram below with the names of the plant parts from Question 3.



How do plants compete for food?

Living things compete for food in a number of ways. Plants make their food by using sunlight, water and other ingredients to make sugars that help them grow. In areas that are crowded with plants, they must compete with other plants for sunlight and water.

Rainforest trees that have their leaves at the top of the canopy (the rainforest 'roof') get the most sunlight. The plants that are lower down get less sunlight, so they are not able to grow as easily, which is why they stay shorter.

 Look at the diagram and answer the questions that follow.



How do plants compete for food? CONT'D

- 1 Which plants in this diagram have received the most sunlight? Colour two in light green, and then describe them below.

- 2 Which plants are the most shaded from sunlight? Colour three in purple, and then describe them below.

- 3 Which plants have photosynthesised less effectively than the tallest trees but more effectively than the lowest bushes? Colour two in dark green, and then describe them below.

- 4 Which living things in this diagram act as decomposers? Colour two in brown, and then describe them below.

- 5 Which plants provide links between trees but are not trees themselves? Colour two in orange, and then describe them below.

- 6 Which plants attach themselves to larger plants in order to grow? Colour two in yellow, and then describe them below.

- 7 Around the edges of the diagram, write the names of living things that might be found in the rainforest (for example, fungi, worms, caterpillar, frog, tree snake, possum, parrot and gibbon). Use arrows to point to the areas these living things would be found.

Bushfires and Australian ecosystems

Australia is a very hot and dry country. In the summer months, bushfires threaten natural habitats all over Australia. Sometimes bushfires start because of lightning strikes or other natural processes. Other times they are deliberately lit and controlled by authorities. Indigenous Australians used controlled burns for thousands of years before Europeans came to Australia.

Although fire destroys some plants in an ecosystem, it encourages other plants to grow. Fire heats up the seed pods of certain trees and plants, such as the banksia plant. When the pressure inside the pod builds up, it bursts open and releases its seeds, which begins a new crop of trees.

Some animals in an ecosystem benefit from bushfires too. Fire encourages new plants to grow and it can reduce animal numbers, which means there is more food and less competition for the surviving animals. Fire also clears old vegetation (plant growth), creating new habitats and shelters for animals.

 **1** In which season do most bushfires occur in Australia? Circle your answer.

autumn winter spring summer

2 What important function do fires have in some parts of the Australian bush? Write your answer below, using the sentence starter to help you.

The heat from bushfires can cause seed pods to _____
_____.

3 Fill in the missing letters to complete these facts about bushfires and ecosystems.

a) A _ _ _ r _ _ _ ia has many bushfires each year.

b) Bushfires can be caused by _ _ _ ghtn _ _ _ strikes.

c) B _ _ nks _ _ _ pl _ _ _ nts need fire so their s _ _ _ ds can germinate, or begin to shoot and gr _ _ w.

d) F _ _ r _ _ _ can destroy the h _ _ b _ _ _ _ _ s of many animal species, but they can also create n _ _ w h _ _ b _ _ t _ _ ts.

e) Eucalyptus trees contain oils that make their leaves _ _ _ rn fiercely.

f) Some plants can r _ _ c _ _ v _ _ r from bushfires, some are destroyed permanently and some n _ _ _ d bushfires to r _ _ pr _ _ d _ _ c _ _ .

Floods and Australian ecosystems

BACKGROUND

habitat (say ‘hab-i-tat’) – the particular place where a group of living things live

marsupial (say ‘marr-soup-ee-al’) – a type of small furry mammal, with a pouch for its young

species (say ‘spee-seas’) – a group of living things that can breed together to produce fertile offspring

Flooding occurs in Australian ecosystems when land that is normally dry becomes covered in water. When there is heavy rain, rivers and creeks may fill up and burst their banks, covering the surrounding land in water. At some times of the year, there may be little rainfall in an area and the ground becomes very hard and dry. When heavy summer rains come, lots of water can fall very quickly. This leads to **run-off**, where water travels across the ground too quickly to be soaked into the soil. A lot of run-off can cause a fast-moving flood, called a **flash flood**.

Floods can be very damaging to some parts of ecosystems. Habitats may be destroyed and animals may drown, upsetting the delicate balance of the ecosystem. But some species benefit from flooding. River red gums need flooding for their seeds to germinate (sprout), and populations of frogs, fish and wetland birds can grow in number after floods.

 **1** Complete the table to show singular and plural forms of these flood words. Write the definition of each word, in your own words.

Singular word	Plural word	Definition
	floods	
rainfall	rainfall	
	creeks	
	shrubs	

- 2

List two negative effects that flooding has on species in ecosystems.
- 3

List two positive effects that flooding has on species in ecosystems.

Check your understanding

Circle the correct answers to these revision questions.

- 1** Plants change the air by
 - a)** absorbing carbon dioxide and releasing carbon.
 - b)** absorbing oxygen and releasing carbon dioxide.
 - c)** absorbing carbon dioxide and releasing oxygen.
- 2** Other than carbon dioxide, what do plants use in the process of photosynthesis?
 - a)** water and sunlight
 - b)** fertiliser
 - c)** oxygen
- 3** Plants are very important in ecosystems because they are
 - a)** consumers.
 - b)** producers
 - c)** directors.
- 4** In which area of the rainforest do plants do the most photosynthesising?
 - a)** the forest floor
 - b)** the canopy
 - c)** the middle
- 5** Which statement is true?
 - a)** Some fires are lit as part of a controlled burn to remove leaf litter.
 - b)** All bushfires begin because of lightning strikes.
 - c)** Australian plants cannot reproduce unless bushfires force their seed pods to burst open.
- 6** What does the word *deliberate* mean?
 - a)** on purpose
 - b)** by accident
 - c)** naturally
- 7** In Australia, floods are most likely to occur in
 - a)** winter.
 - b)** summer.
 - c)** spring.
- 8** The name we give to a fast-moving flood caused by run-off is
 - a)** a flash flood.
 - b)** high tide.
 - c)** an annual flood.

How do animals in ecosystems compete for food?

All animals need to eat food. Plant-eaters may **forage** (search) for food among the plants, picking fruits, berries and nuts from trees and bushes. Others **graze** on crops planted by humans or grass. Most meat-eaters **hunt** and kill other animals.

Sometimes animals need to get their share of food by fighting each other. This is called **competition**. Sometimes it might be two members of the same species that are fighting – for example, two young monkeys might fight over a piece of fruit. Sometimes it is two different kinds of animals that fight for food – for example, a hyena might try to take some food from a lion.

-  **1** Circle the word that relates to the way in which animals fight for food.

communication	competition	accommodation
---------------	-------------	---------------

- 2** Draw a line between each sentence starter and the most appropriate sentence ending.

- | | |
|-------------------|-------------------------------------|
| a) Foraging means | chasing and killing another animal. |
| b) Grazing means | feeding on grass or plants. |
| c) Hunting means | searching for wild food. |

- 3** Give two examples for each of the following.

- | | | |
|------------------------|-------|-------|
| a) animals that forage | _____ | _____ |
| b) animals that graze | _____ | _____ |
| c) animals that hunt | _____ | _____ |

- 4** What might help an animal win a contest for food? Underline all your choices.

speed	strength	sleepiness	having babies to feed
sharp claws	dancing ability	illness	being overweight
having an injury	poor eyesight	climbing ability	sharp teeth
intelligence	long fur	being healthy	camouflage

What are predators?

Predators are living things that hunt and kill other living things for food. Their target is called the **prey**.

Big cats, such as lions, tigers, leopards and cheetahs, are among the world’s most successful predators. They can run very fast and have sharp teeth and claws. They are also able to hide themselves in their environment because their fur is patterned to make them difficult to see. This is called **camouflage**. It is a common feature amongst both predators and prey.

 **1** What is a predator? Fill in the gaps below, and then write out your new definition.

A predator is an _____ that hunts and _____
other _____ to obtain food.

2 Fill in the gaps in the table to show that you understand nouns and verbs.

Singular noun	Plural noun	Verb
	predators	predating
prey		preying
hunter		
	targets	

3 Complete the following sentences.

a) Camouflage helps predators because _____
_____.

b) Camouflage helps prey because _____
_____.

What are predators? CONT'D

- 4 a) Separate the letter chain into individual living things, then list each one in the table as predator or prey.
- b) Circle any that could be both predator and prey.

cheetahharewildebeestwolfgoatantelopefoxmouselambzebrameerkatrabbitcobra
scorpionspidereagle

Predators	Prey

- 5 Which of these animals are the fiercest predators? Write your top four choices, and explain why you chose them.

tiger rabbit leopard spider monkey crocodile frog sheep
wolf lizard guinea pig human eagle chimpanzee

Predator 1: _____ Explanation: _____

Predator 2: _____ Explanation: _____

Predator 3: _____ Explanation: _____

Predator 4: _____ Explanation: _____

Relationships

Living things in an ecosystem interact in different ways. Some living things have learned to depend on each other to survive.

Collaboration

Animals of the same species may live in groups, rather than on their own. A group of animals from the same species is called a **population**. As a population, animals can share the jobs of finding food and water, protecting each other from predators, raising their young and finding shelter. This is called **collaboration**. It means ‘working together’.

Symbiotic relationships

Certain species develop a relationship with a completely different species that benefits one or both animals. This is called a **symbiotic relationship**. When the symbiotic relationship benefits both species it is called **mutualism**. An example is the clownfish, which lives amongst the poisonous tentacles of a sea anemone. The clownfish protects the anemone from the bigger fish that eat anemones. In return, the clownfish is protected from predators by the anemone’s stinging tentacles. The clownfish is not hurt because it has a special mucus on its skin that protects it from the sting.

-  1 Words from the text have been broken into syllables and the letters have been scrambled. Unscramble the syllables and put the words back together. Write each word next to its definition.

Scrambled syllables	Word	Definition
ed / pedn		It means ‘rely’.
mys / ib / o / cit		It describes a relationship that benefits one or both species.
rsu / viev		It means ‘stay alive’.
co / ball / o / ra / toni		It means the process of working together.
op / up / la / iton		It means a group of animals from the same species.

Relationships CONT'D

Parasitic relationships

Some living things have a relationship where one is a **host** and the other is a **parasite**. Fleas are common parasites. Fleas are found on the bodies of furry animals, including rats and dogs. These parasites benefit by feeding on the blood of the host. The host gains nothing from the relationship at all. The host may feel itchy and can become weak, miserable and even sick from too many fleas. We call too many parasites in one place an **infestation**.

Parasites such as fleas and ticks feed on the blood of many types of mammals. Other parasites may feed on flesh (skin tissue), hair proteins, sweat or salts.

- 2 Write the living things from the word bank into the table, showing which ones are parasites on which hosts. An example has been done for you. Hint: Some living things may host more than one type of parasite, and some parasites are found on more than one type of host.

Hosts	Parasites
dogs	ticks, sandflies

Word bank

fleas

birds

humans

lice

cats

fish

barnacles

dogs

mosquitoes

sandflies

ticks

- 3 Name four things that parasites might eat.
-
-

Drought and Australian ecosystems

Drought describes a time of abnormally low rainfall in a certain area, when there is not enough water to meet normal needs. Australia is often affected by drought. Sometimes a drought can last for many years.

If water supplies in an ecosystem are low for a long time, plants wither and die. This causes **food shortages** for all the living things that rely on those plants for food, and loss of shelter for some animals. Eventually, the animals that do not have enough food stop breeding and may die out. The food web of an ecosystem can be harmed in this way. In places where drought has gone on for a long time, some ecosystems never recover.

 **1** Solve the puzzles to complete the list of words about drought.

- | | |
|--------------|--|
| a) _ _ _ _ _ | This word means 'a time when there is unusually low rainfall' and begins with the letter that comes after 'c' in the alphabet. |
| b) _ _ _ _ _ | This name of a country has 'u' as its second letter. |
| c) _ _ _ | This word means the opposite of 'live'. |
| d) _ _ _ _ _ | This word means 'a place to get protection from the weather'. |
| e) _ _ _ _ _ | This word has 'ee' in it and means 'reproducing'. |
| f) _ _ _ _ _ | This word means 'a particular type of living creature' and starts and ends with 's'. |

2 Complete the passage by writing your answers from Question 1 into the correct spaces.

A time of unusually low rainfall in a place is called a _____. Some parts of _____ are frequently affected by drought. Plants can _____ if a drought lasts a long time. When plants are lost, animals that need plants for food and _____ are affected. If they do not have enough food, animals may stop _____. This makes it more likely that the _____ will become endangered or die out.

Check your understanding

Circle the correct answers to these revision questions.

- 1 What does camouflage enable some animals to do?
 - a) run faster
 - b) hide in their habitat
 - c) sharpen their claws
- 2 Which term means 'searching for food among plants'?
 - a) foraging
 - b) competition
 - c) predating
- 3 What vegetation do animals such as rabbit and deer graze on?
 - a) snakes and fish
 - b) tall trees
 - c) crops planted by humans and grass
- 4 What is a population?
 - a) different groups of animals from a variety of species
 - b) different types of animals that are all predators
 - c) a group of animals from the same species
- 5 What is the relationship between a sea anemone and a clownfish called?
 - a) a biological relationship
 - b) a symbiotic relationship
 - c) a bionic relationship
- 6 Which two parasites feed on the blood of mammals?
 - a) ticks and fleas
 - b) nits and barnacles
 - c) dogs and birds
- 7 Food shortages occur in some ecosystems in Australia due to
 - a) doubt.
 - b) draughts.
 - c) drought.
- 8 What might animals stop doing during drought, which may lead to their species dying out?
 - a) breeding
 - b) making shelters
 - c) growing

How do plants reproduce?

Background

fertilisation say ('furr-til-iz-ay-shen') – the merging (joining) of the male and female reproductive cells

All plants and animals reproduce. Reproduction may be **asexual** (without sex) or **sexual**.

A plant that reproduces asexually produces a new plant by itself. The new plant is genetically identical to the parent plant. This is because it has the same **genes** as the parent plant. Genes are information carried in a living thing's cells that determines some of its characteristics.

A plant that reproduces sexually produces a new plant through **fertilisation**, either by itself or with another plant. The new plant is genetically different to both the parent plants, because it has a combination of the genes of its parents.

 **1** Fill in the missing vowels (a, e, i, o, u) to complete the sentences.

- S _ x _ _ l r _ pr _ d _ ct _ _ n involves f _ rt _ l _ s _ t _ _ n, which is the joining of a m _ l _ reproductive cell and a f _ m _ l _ reproductive cell.
- _ s _ x _ _ l r _ pr _ d _ ct _ _ n involves only one plant, which may form new tiny plants called plantlets or bulbs.

Pollination

Flowering plants reproduce through a process called **pollination**. Flowers contain the reproductive organs of a plant. Some plants have flowers with both male and female parts, while other plants have flowers with only male parts or only female parts.



The stamen is the male part of the flower. The pistil is the female part of the flower. The butterfly transfers pollen grains from the stamen to the pistil.

How do plants reproduce? CONT'D

2 Fill in the blanks with the correct word from the word bank.

	Word bank
Some flowers contain tiny yellow grains called _____, which are male cells. This pollen is carried by bees, _____ or other insects to the female parts of the same or another _____.	butterflies flower pollen
The male and female parts _____ together in fertilisation. In this way, insects help to _____ flowers.	fertilise join
Fertilisation produces _____, which can develop into new _____. The plant develops these seeds inside its _____ and releases the fruit at certain times of the year.	fruit seeds plants
When the fruit _____ to the ground and _____, the seeds are released and go into the soil. When birds eat the fruit or seeds of a plant, the seeds _____ through the bird's digestive system and come out in its droppings. In this way, the seeds of plants can travel long distances from their parent plant.	rots travel falls

- 3 What is pollen? Circle the correct answer.
- plant seeds flowers reproductive material fruit
- 4 Where are the seeds of a flowering plant? Circle the correct answer.
- in its pollen in its fruit in its flowers in its stem
- 5 Name at least four types of animals that help flowers pollinate and spread their seeds.
-

How do animals reproduce?

Background

fertilisation (say ‘furr-til-iz-ay-shen’) – the merging (joining) of the male and female reproductive cells

species (say ‘spee-seas’) – a group of living things that can breed together to produce fertile offspring

Animals come in all shapes and sizes. Most of them reproduce through **sexual reproduction**, but they **breed** and have their young in many different ways.

Two partners

Sexual reproduction requires two partners of the opposite sex. Because of this, all species that reproduce sexually come in two different forms: males and females. An important principle of sexual reproduction is that the male and female must be of the same species if they are to successfully reproduce.

In some species, males and females look very different. In other species, the differences are not so obvious.

-  1
- Fill in the gaps in the table, naming the female and male forms of these animals. Leave the final column blank for now.

Animal	Female form	Male form	
human	w_____	m_____	
c_____	hen	r_____	
elephant	b_____	c_____	
h_____	m_____	stallion	
s_____	ewe	r_____	

- 2 a)
- Name two types of animals that have male and female forms that look quite different.

- b)
- Why do you think the male and female forms look different?

How do animals reproduce? CONT'D

Fertilisation and offspring

Many animals that reproduce sexually **mate**. This means that the male fertilises the egg of the female inside the female, as in human reproduction. Other animals, such as frogs and some fish, fertilise the female’s eggs outside her body, after they have been laid.

The single fertilised cell begins to divide and a foetus begins to grow. The foetus spends time growing inside the female – called **gestation** in most animals or **pregnancy** in humans – or it may grow inside an egg that has been laid. After a certain amount of time, the new offspring is either born or hatches from the egg.

Mammals give birth to live offspring. Reptiles, birds, fish and insects hatch from eggs.

- 3
- a) Fill in the second column of the table by writing two examples of each type of animal.

b) Fill in the third column of the table by writing ‘egg’ (if the offspring usually hatches from an egg) or ‘live baby’ (if the offspring is born from the mother).

Animal type	Two examples	Egg or live baby	
mammals			
reptiles			
birds			
fish			
insects			

- 4
- Go back to Question 1. Write the heading ‘Offspring’ in the fourth column of the table and write in the name given for each type of offspring.

DNA, chromosomes and genes

BACKGROUND

nucleus (say 'nu-kle-us') – the round structure at the centre of a cell, containing genetic information that controls what the cell does

A new life begins as just one **cell**. All the information for growing a living thing is found inside the nucleus of this cell. For most animals, this cell is a fertilised egg, containing **genetic information** from two parents. This genetic information is stored in DNA, in genes and chromosomes.

DNA

- The letters DNA stand for deoxyribonucleic acid (say de-ox-ee-rye-bo-new-clee-ick ass-id). When people talk about DNA, they usually mean the information that DNA contains, not the acid itself.
- DNA is found inside the cells of nearly all living things.
- DNA is like a set of instructions for building and operating a living thing. It provides the information to make the **proteins** that make up cells, as well as instructions as to how the cells form, what shape they will be, what things they can do and so on.

Chromosomes and genes

- Chromosomes are tiny threads made of DNA. Genes are short pieces of these threads.
- Each chromosome consists of thousands of different genes.
- Every gene carries information about a specific characteristic.
- Chromosomes are found inside the **nucleus** of cells. In each cell, the genetic information is exactly the same, containing the same instructions.
- Chromosomes come in pairs. Humans have 23 pairs of chromosomes.
- One chromosome of every pair comes from the mother and one in every pair comes from the father.

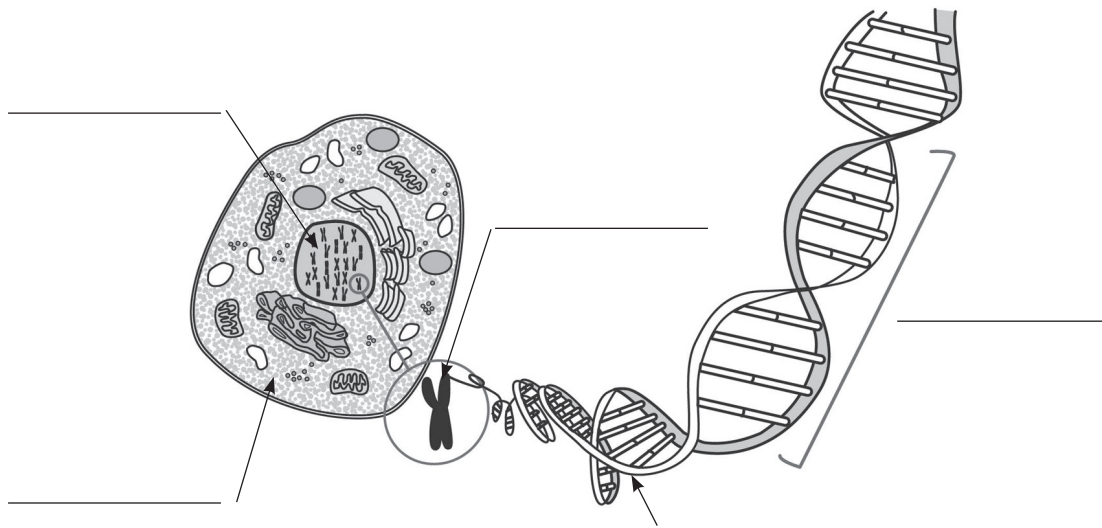
DNA, chromosomes and genes CONT'D

1 Are the following statements true or false?

- | | | |
|---|------|-------|
| a) The letters DNA stand for dioxide ribbons and nuclear acids. | True | False |
| b) DNA contains information to make proteins in particular ways. | True | False |
| c) DNA is the genetic material of living things. | True | False |
| d) Our parents can decide on and control what material goes into our genes. | True | False |

2 Look at the diagram. Label the features listed in the box.

chromosome nucleus cell gene DNA



3 Rearrange these sentence parts to make sentences about chromosomes. Write the correct sentences on the lines provided.

- a) tiny threads DNA Chromosomes are that are made of

- b) the genetic material the same instructions contains exactly In each cell,

Adaptation

BACKGROUND

habitat (say 'hab-i-tat') – the particular place where a group of living things live

marsupial (say 'marr-soup-ee-al') – a type of small furry mammal, with a pouch for its young

species (say 'spee-seas') – a group of living things that can breed together to produce fertile offspring

Adaptation is the name that scientists give to the way that living things **change** and become **better suited** to their habitats.

Offspring are not identical to either of their parents. Each new generation of a **species** is a tiny bit different from the one before. The **theory of evolution** says that these tiny differences are not noticeable from generation to generation, but over thousands or millions of years these small differences can completely and obviously change the way a plant or animal looks or functions. Adaptation results in new species and new subspecies. In this way it contributes to the **biodiversity**, or biological diversity, of life on Earth.

 **1** Explain why you think offspring are not identical to either of their parents.

The process of adaptation happens because of small **variations** within a species. Variations that give some individuals a better chance of survival than other individuals in their species lead to what we call 'survival of the fittest', or **natural selection**. An example of a variation that might give a better chance of survival is a different fur colour, which might make it easier for a small animal to hide from predators.

Because the changed individuals are more likely to survive, they are more likely to breed and produce offspring. Their offspring are more likely to have the changed feature, or **trait**. Over time, this can result in a complete change to all the individuals in a species. The new species trait is called the **adaptive trait**, or the adaptation.

Adaptation CONT'D

2 Add the missing vowels (a, e, i, o, u) to complete the following words. Hint: All the words have been used on the previous page, in this order.

- a) dpttn _____
- b) spcs _____
- c) vrtns _____
- d) srvvl _____
- e) prdtrs _____
- f) ffsprng _____
- g) trt _____

The circumstances within an ecosystem can determine whether a small variation within a species is of benefit or not. For example, within one species of small marsupial in a woodland there are slight variations in fur colour, but most have quite dark fur. Over time, the climate of the forest changes and there is more and more snow. Marsupials that have light-coloured fur are able to escape from predators more often than marsupials with darker fur. So more marsupials with light-coloured fur survive to reproduce, and they reproduce offspring that are more likely to have lighter fur. Over many generations, the species of marsupial in the woodland changes to become white-furred marsupials.

3 Are the following statements about the example adaptation true or false?

- a) Marsupials with light-coloured fur can run faster in the snow.

TrueFalse
- b) Marsupials with light-coloured fur can hide better in the snow.

TrueFalse
- c) Adaptation resulted in two species of marsupials, one light and one dark.

TrueFalse
- d) The change happened quickly but it wasn't permanent.

TrueFalse
- e) The circumstances in the ecosystem determined which individuals were best suited to survive.

TrueFalse

Natural selection

BACKGROUND

genes (say 'jeans') – information carried in a living thing's cells that determines some of its characteristics

Natural selection is the term we give to the way in which a particular animal or plant survives to breed. Natural selection results in species **adaptation**. Evolutionary scientists call this process **evolution by natural selection**.

Some animals within a species – a group of rabbits, for example – will be healthier, stronger and quicker than others. If this group of rabbits were to be chased by a wolf or a fox, the 'fittest' ones would survive to breed because they could run faster and for longer periods of time. These better-adapted rabbits would eventually have babies, and most of the young rabbits might also inherit their parents' strength and fitness. Over time, the less fit rabbits would die out and the stronger, fitter, faster rabbits would survive. Nature is 'selecting' those rabbits to be the ones that pass their genes down to a new generation.

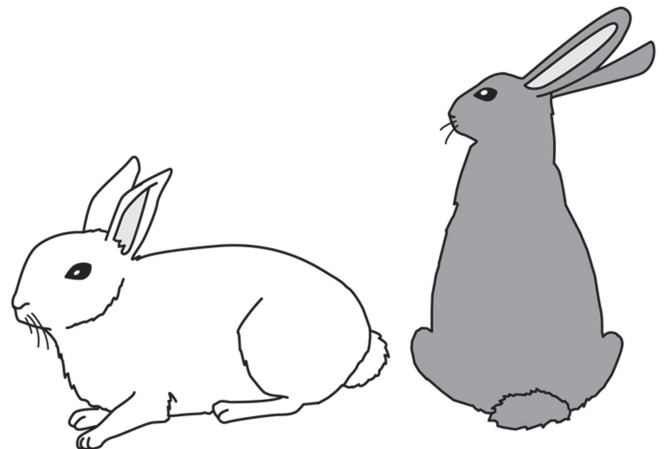
 **1** Complete this sentence, using your own words.

The word 'fittest' means _____.

2 Use just one word from the box to fill the spaces, and complete the passage.

diseases	babies	parents	rabbits
----------	--------	---------	---------

The _____ of the 'fittest' creatures will be more likely to survive to have _____ of their own. It is highly likely that those _____ will inherit the 'stronger' features of their parents. They will also probably pass those genes on to their own _____. There will be less and less individuals with 'weaker' features, as those with 'stronger' features dominate. The 'weaker' individuals will die out and only the 'fittest' will remain.



Natural selection CONT'D

- 3 Which of these rabbits show features that would most likely be 'selected' and help a group of rabbits survive? Circle your choices.
- a) this small, weak male rabbit loses contests with stronger male rabbits to mate with a female
 - b) this rabbit with white fur is able to hide better in its snowy habitat than a rabbit with dark fur
 - c) this pet rabbit is chased by a dog and is killed
 - d) this rabbit is able to run faster than another when it is being chased by a wolf
 - e) this rabbit becomes blind
 - f) this rabbit is healthier than others and is able to survive through a drought
 - g) this rabbit fails to build itself a burrow before a snowy winter sets in
 - h) this rabbit population has larger teeth that allow the rabbits to chew thicker, more nutritious grass than other smaller-toothed rabbits
 - i) this male rabbit has shorter fur that allows it to stay cooler in its hot desert habitat than furrier rabbits
 - j) this group of rabbits has chosen to live in an area that is a popular hunting ground where people come to shoot rabbits
 - k) this rabbit's family built a better network of burrows than another group
 - l) this pet rabbit refuses to eat vegetables or plants of any kind
 - m) this rabbit has its burrows in an area where foxes begin to roam
 - n) this rabbit has immunity to a disease that kills other rabbits
 - o) this rabbit has fur patterns that make it less attractive to mating partners than other rabbits

Check your understanding

Circle the correct answers to these revision questions.

- 1** What process is occurring when insects transfer pollen from one flower to another?
 - a)** pollination
 - b)** digestion
 - c)** growth
- 2** What is one way that fruit helps release the seeds of plants?
 - a)** the seeds are carried on the legs of insects
 - b)** the fruit rots and the seeds go into the soil
 - c)** the seeds fly away in the wind
- 3** Which of these categories of living things do not lay eggs?
 - a)** humans
 - b)** frogs
 - c)** fish
- 4** What occurs during mating in mammals?
 - a)** animals kill each other
 - b)** babies from two different species develop
 - c)** the egg or eggs of the female are fertilised by a male
- 5** What is the purpose of DNA in cells?
 - a)** it contains the information required to make proteins, which form body cells
 - b)** it helps us to digest food
 - c)** it makes protein
- 6** Which statement is correct?
 - a)** Chromosomes are found on genes.
 - b)** Chromosomes and genes are found in DNA.
 - c)** Genes are found on chromosomes.
- 7** How many chromosomes do humans have in total?
 - a)** 46 (or 23 pairs)
 - b)** 23
 - c)** each person has a different number
- 8** What does the process of natural selection lead to?
 - a)** It helps make animals get more DNA.
 - b)** The 'fittest' survive and pass their genes on to the next generation.
 - c)** It ensures that rabbits can survive better than their parents did.

The human species

BACKGROUND

mammals (say ‘ma-mels’) – warm-blooded animals with backbones that feed their young with milk

Humans are an important part of the biodiversity of ecosystems.

Human reproduction

Humans reproduce in the same way as other mammals. After sexual intercourse, a single **sperm cell** fertilises the female’s **ovum** (egg). Usually, one egg is released from the female’s ovaries every month. If the male sperm cell reaches the egg at just the right time, the sperm goes into the egg and the two cells join together and begin multiplying. The cells continue to grow for nine months inside the female’s womb. The human baby is born after passing through the female’s vagina (birth canal).

-  **1** Underline the correct word within the brackets to complete these sentences.
- a)** Human reproduction is (similar / different) to the way other mammals reproduce.
 - b)** A (male / female) sperm cell fertilises the egg of a (male / female).
 - c)** The physical activity that enables the egg and the sperm to come into contact is called sexual (intercourse / joining).
 - d)** An egg has been fertilised if a (sperm cell / ovum) has gone into it.
 - e)** The multiplying cells eventually become an (embryo / womb).
 - f)** The baby travels through the (ovaries / vagina) during birth.

- 2** Complete the table by adding the missing letters.

Singular noun	Plural noun	Adjective
cell	_____	_____ ular
female	_____	_____
mammal	_____	_____ ian
sex	_____ es	_____ ual
ovary	_____	_____ ian
embryo	_____	_____ nic

The human species CONT'D

Human babies

Just like young mammals, human babies have special needs that they look to their parents to provide. They include the same things that other mammals need – food, water, shelter, warmth and safety. Human babies also have emotional and social needs. Humans usually make a home for their young where they can be kept safe and where they feel part of a family or community group.

- 3 List the similarities between human offspring and the offspring of other mammals.

- 4 List the differences between human offspring and the offspring of some other mammal species.

Human adaptations

Most scientists believe that humans have changed over millions of years. **Adaptation** is the name given to describe the way that living things change and become better suited to their habitats. For example, human skin tones and hair colour have adapted to suit living in different climates.

Like other animals, humans are still changing. These changes are very tiny, but they will be very obvious over thousands or millions of years.

- 5 Fill in the vowels to complete this sentence.

H _ m _ ns b _ l _ ng to a sp _ c _ _s called *Homo sapiens*.

- 6 What features do you think will help the human species survive the next 10 000 years? List each feature in your workbook, and explain why you think it will be important. Then draw a picture of your new human species.

Human effects on ecosystems

Humans are part of many different types of ecosystems, just like other animal species. The actions of humans in an ecosystem, however, can negatively affect an ecosystem’s delicate balance. Food chains and food webs can be affected in ways that threaten the survival of some species.

-  1 Complete the table, giving at least two examples of short-term effects and two examples of long-term effects of each human activity.

Human activity	Short-term effects	Long-term effects
building cities (think about: clearing land, human population increases, waste products and pollution)		
growing crops (think about: clearing land, using fertilisers, introducing new plant species)		
burning fossil fuels for energy (think about: mining, pollution, building power plants)		

Human effects on ecosystems CONT'D

Introduced species

Human activities can introduce species into ecosystems where they normally wouldn't appear. The introduced species are called non-native species. They can affect the delicate balance of an ecosystem and have a negative effect on populations of native species.

2 Circle the animal species that have been introduced into Australian ecosystems.

cane toads	wallabies	sharks
rabbits	camels	brumbies
feral cats	brown bears	funnel-web spiders
foxes	crocodiles	goats

Feral cats can be found in all types of habitats across Australia. These predators disturb ecosystems by killing birds and small mammals and causing the extinction (dying out) of some animal species that live on the ground.

3 Using your dictionary, write a definition for the word *feral*.

4 How do you think cats were introduced into Australian ecosystems?

5 Give two possible effects on an ecosystem if a population of animals is wiped out by an introduced species. Use the example of a feral cat wiping out a population of bandicoots in a woodland ecosystem.

Check your understanding

Circle the correct answers to these revision questions.

- 1** What are the reproductive cells of the male and female called in mammals?
 - a)** sperm and womb
 - b)** sperm cell and ovum
 - c)** egg and fluid
- 2** Where inside the female does a human baby grow?
 - a)** the embryo
 - b)** the birth canal
 - c)** the womb
- 3** The young offspring of mammals need
 - a)** food, water, shelter, warmth and safety.
 - b)** only food and water.
 - c)** food, water, shelter and darkness.
- 4** The human species is called
 - a)** *Homo erectus*.
 - b)** *Homo sapiens*.
 - c)** *Homo habilis*.
- 5** Which of the following is an example of human adaptation?
 - a)** building cities
 - b)** skin and hair colour changes
 - c)** wearing clothes
- 6** Some human activities that can negatively affect ecosystems are
 - a)** growing crops and mining.
 - b)** fishing and hunting.
 - c)** both of the above.
- 7** Humans are
 - a)** not part of ecosystems.
 - b)** part of many types of ecosystems.
 - c)** in control of everything that happens in an ecosystem.
- 8** Introduced species are also called
 - a)** native species.
 - b)** non-native species.
 - c)** Australian species.

1 Living things and ecosystems

What is an ecosystems? **1 a** the living and non-living things in a certain place and all their interactions **b** a set of connected things that form a whole **c** act in ways that affect one another **d** a living thing **e** all the living things that live in the same place and depend on one another **f** the variety of different lifeforms in a particular place or ecosystem. **2** Sample answers: water birds eat frogs, fish eat algae, frogs lay their eggs under rocks, water plants use sunlight to make their own food. **3** Sample answers: eat the food of another species, causing them to move away or die out; drown animals; plants can't grow without water, affecting everything that eats the plants. **More about ecosystems** **1** food, eat, diagram, water, creek, ground, air, light, cool, homes, safety, habitat. **2 a** use energy from sunlight to create their own food. **b** eat plants and release the energy in the food. **c** break down animal and plant remains into small parts. **3** Sample answers: **a** rainforest **b** meadow **c** plants, fungi, flowers. **4** Sample answers: **b** cacti, grasses **c** banksias, eucalyptus trees **d** water lilies, lotuses **e** ferns, vines. **What is a habitat?** **1** rabbit: fresh grass to graze on, underground burrows; clownfish: coral reef, the protection of a sea anemone; spider monkey: fruit trees, a protective family group; koala: eucalyptus leaves, sharp claws and teeth. **2 a** shelter **b** food **c** clean air **d** water. **3** Answers will vary. **Ecosystems and habitats** **1** ecosystem, habitat, community, population. **2 a** ecosystem, habitat **b** ecosystem, community. **3–4** Answers will vary. **Changing behaviour** **1** adaptation. **2** adapt. **3 a. 4** drink more water, lie in the shade, limit movement at certain times of the day. **Relationships in ecosystems** **1** Sample answer: Mutualism describes a relationship between living things where both organisms benefit. **2 a** false **b** false **c** true **d** false. **3** skin, smaller, algae, covered, mutualism, benefit, food. **4** organism, algae, skin, relationship. **5 a** yes **b** no. **Competition in ecosystems** **1 a** peace and harmony **b** in order to survive **c** help them to reproduce. **2 a** & c. **3** Sample answers: **a** food, water, shelter, mating partners **b** sunlight, growing space, water, nutrients. **4 a** Ecosystems are not always peaceful. **b** Living things compete in different ways. **c** Animals compete for mating partners. **d** Plants compete with each other for sunlight. **e** Plants need growing space and nutrients from the soil. **5** Answers will vary. **Check your understanding** **1 c. 2 a. 3 c. 4 b. 5 b. 6 b. 7 a. 8 c.**

2 Food and ecosystems

Producers, consumers and decomposers

1 b Sample answer: No. Many of the decomposers are microscopic organisms, which means they can't be seen with the naked eye. **2** Answers will vary. **3** Sample answers: **a** coming from living things **b** not coming from living things. **What is a producer?** **1 a** Humans are consumers. **b** Producers use inorganic ingredients to produce their food. **c** The word produce means 'to make'. **d** The word inorganic means 'coming from non-living things'. **2** Answers will vary. Make sure that students list species, rather than general animal types. **3** wheat, cherries, watermelon, corn, beans, tomatoes, onions, potatoes, bananas, apples, carrots, grapes, sesame seeds, olives, oats, rice, celery. **4 a** sugar cane and sugar beet **b** peanuts **c** coffee beans (from trees) **d** tea leaves (from tea bushes) **e** oranges from fruit trees **f** soya bean plants **g** corn plants **h** olive trees **i** grapevines **j** wheat crops. **5** Answers will vary. **What is a consumer?** **1** Answers will vary. **2 a** consumer **b** producer **c** consumer **d** consumer **e** producer. **3 a** tiger, brown bear, dingo **b** caterpillar, snail, zebra, mouse, rabbit **c** tiger: carnivore, brown bear: omnivore, dingo: omnivore. **4 a** blue whale, crocodile, lion **b** humans. **What is a decomposer?** **1** termite bacteria, mould, toadstool, maggot, worm. **2** de/com/pose, de/com/pos/ers, de/com/pos/i/tion. **3** Decomposers, organisms, smaller, dung, dead, plants, decompose, nutrients, soil, recycled, young, animals. **4** Sample answers: **a** fungi **b** termites **c** dung flies **d** fruit flies **e** maggots **f** slugs **g** lactobacteria. **5** fruit, vegetables, decay, smelly, Decomposers, warm, dark, bacteria, mould, fungi, break, nutrients, chemicals, absorb, Manure, rotten, meat, carcasses, carbon, ecosystem. **Food chains** **1** Food chains should be written in the following order, with the arrows pointing to the right: **c** (grass rabbit ferret) **d** (plankton salmon brown bear) **e** (seaweed small fish tuna cormorant) **f** (insect rat raccoon lynx) **g** (fly frog woodpecker wolf) **h** (grain fieldmouse weasel cougar) **i** (worm prawn whiting jewfish shark) **j** (fruit grub monkey hyena) **k** (plankton mussel starfish sea snail crab). **Food webs** **1 a** snake **b** spider, cricket **c** plants, weeds **d** Sample answer: Australian rural ecosystem. **2 a** food, sources, killing, smaller, eggs, plants, illness, attack. **b** Answers will vary. **3** Answers will vary. **Check your understanding** **1 c. 2 b. 3 a. 4 b. 5 c. 6 a. 7 a. 8 c.**

3 Plants in ecosystems

The importance of plants in ecosystems 1 **a** carbon dioxide **b** leaves **c** photosynthesis **d** glucose and oxygen **e** oxygen **f** respiration (breathing) **g** carbon **h** carbon dioxide **i** water **j** oxygen and hydrogen. 2 Answers will vary. 3 Sample answers: Plants are producers, which means they are a source of food for other living things. Plants also take in carbon dioxide and release oxygen. **Photosynthesis and ecosystems** 1 carbon dioxide, ecosystem, energy, glucose, nutrients, photosynthesis, water. 2 **a** glucose or food **b** Sample answers: photograph, photocopy, photochemical. 3 leaves, flowers, fruit, stalk, roots. **How do plants compete for food?** 1 the tallest trees. 2 the lowest shrubs. 3 the second-tallest trees. 4 fungi, moulds and other ground-based organisms. 5 vines. 6 staghorns, elkhorns and similar plants. 7 Answers will vary. **Bushfires and Australian ecosystems** 1 summer. 2 Sample answer: burst open and release seeds, so a new crop of trees may start to grow. 3 **a** Australia **b** lightning **c** Banksia, plants, seeds, grow **d** Fires, habitats, new habitats **e** burn **f** recover, need, reproduce. **Floods and Australian ecosystems** 1 flood, creek, shrub. Definitions will vary. 2 Sample answers: habitats destroyed, animals drowned. 3 Sample answers: gum trees can germinate, some animal numbers increase. **Check your understanding** 1 **c**. 2 **a**. 3 **b**. 4 **b**. 5 **a**. 6 **a**. 7 **b**. 8 **a**.

4 Animals in ecosystems

How do animals in ecosystems compete for food? 1 competition. 2 **a** searching for wild food. **b** feeding on grass or plants. **c** chasing and killing another animal. 3 Answers will vary. 4 speed, strength, sharp claws, climbing ability, sharp teeth, intelligence, being healthy, camouflage. **What are predators?** 1 animal, kills, animals. 2 predator, prey, hunters, hunting (or hunt), target, targeting (or target). 3 Sample answers: **a** they can sneak up on their prey **b** they can hide from predators. 4 **a** Sample answers: Predators: cheetah, wolf, fox, cobra, scorpion, spider, eagle. Prey: hare, wildebeest, goat, antelope, mouse, lamb, zebra, meerkat, rabbit. **b** fox, meerkat, cobra, scorpion, spider. 5 Answers will vary. **Relationships** 1 depend, symbiotic, survive, collaboration, population. 2 Sample answers: birds: lice; humans: lice, mosquitoes, sandflies, fleas, ticks; fish: barnacles. 3 Sample answers: blood, flesh, hair proteins, sweat, salts. **Drought and Australian ecosystems**

1 **a** drought **b** Australia **c** die **d** shelter **e** breeding **f** species. 2 drought, Australia, die, shelter, breeding, species. **Check your understanding** 1 **b**. 2 **a**. 3 **c**. 4 **c**. 5 **b**. 6 **a**. 7 **c**. 8 **a**.

5 Plant and animal biodiversity

How do plants reproduce? 1 **a** Sexual reproduction, fertilisation, male, female **b** Asexual reproduction. 2 pollen, butterflies, flower, join, fertilise, seeds, plants, fruit, falls, rots, travel. 3 reproductive material. 4 in its fruit. 5 Sample answers: butterflies, bees, other insects, birds. **How do animals reproduce?** 1 woman, man, chicken, rooster, bull, cow, horse, mare, sheep, ram. 2 **a** Answers will vary. **b** Sample answer: it makes it easier to identify a mate. 3 **a** Answers will vary. **b** live baby, egg, egg, egg, egg. 4 baby, chick, calf, foal, lamb. **DNA, chromosomes and genes** 1 **a** false **b** true **c** true **d** false. 3 **a** Chromosomes are tiny threads that are made of DNA. **b** In each cell, the genetic material contains exactly the same instructions. **Adaptation** 1 Sample answer: Their features are a unique combination of both parents. 2 **a** adaptation **b** species **c** variations **d** survival **e** predators **f** offspring **g** trait. 3 **a** false **b** true **c** false **d** false **e** true. **Natural selection** 1 Sample answer: the best able to survive. 2 babies. 3 **b**, **d**, **f**, **h**, **i**, **k**, **n**. **Check your understanding** 1 **a**. 2 **b**. 3 **a**. 4 **c**. 5 **a**. 6 **c**. 7 **a**. 8 **b**.

6 Humans and ecosystems

The human species 1 **a** similar **b** male, female **c** intercourse **d** sperm cell **e** embryo **f** vagina. 2 cells, cellular, females, female, mammals, mammalian, sexes, sexual, ovaries, ovarian, embryos, embryonic. 3 Sample answers: they need food, water, shelter and safety. 4 Answers will vary. 5 Humans, belong, species. 6 Answers will vary. **Human effects on ecosystems** 1 Answers will vary. 2 cane toads, rabbits, camels, brumbies, feral cats, foxes, goats. 3 Sample answer: in a wild state, especially after escaping from captivity. 4 Sample answer: Cats were brought from other countries as pets or to be used on farms to control pests such as mice. Cats escaped or were let loose and became feral. 5 Sample answer: Animals that prey on the bandicoot will be affected (numbers may fall). Animals that are preyed on by the bandicoot will be affected (numbers may rise). **Check your understanding** 1 **b**. 2 **c**. 3 **a**. 4 **b**. 5 **b**. 6 **c**. 7 **b**. 8 **b**.

Your Achieve! Science CD-ROM

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Quality ready-to-use resources

Each book in the **Achieve!** series has been designed and written for secondary students who have low-level literacy skills and require modified classroom activities to fully participate in the curriculum. Students will experience success as they work alongside their classmates on the same subjects with material that is tailored for their literacy needs.

Key features

- ✓ Mature-look page layout and illustrations
- ✓ Topics that will interest teenagers
- ✓ An emphasis on literacy skills
- ✓ Revision sheet at the end of each unit
- ✓ Background boxes define key terms and give a pronunciation guide

The work sheets in this series are ideal for group or individual study and are easy for classroom helpers to implement in one-on-one lessons. You can also use the photocopiable pages for revision, assessment and home study.

Achieve! Science Ecosystems

While also suitable for the regular classroom, each of the **Achieve! Science** titles has been written at a level that is appropriate for secondary school students who, because of poor literacy skills, may struggle with regular classroom materials.

Ecosystems covers a variety of topics, including how living things interact in ecosystems, food chains and food webs, habitats, animal and plant reproduction, adaptation, evolution by natural selection, and the effects of floods, drought and bushfires on Australian ecosystems.

This book is divided into six units, all of which contain photocopiable Activity sheets and conclude with a Revision sheet to test understanding of the unit content. Varied activities – including labelling diagrams, completing tables, multiple-choice questions, cloze activities and word games – allow students to build their scientific vocabulary and understanding of key concepts.

The CD-ROM contains all the pages that are featured in the book, as well as a file of selected illustrations that can be used for other work sheets. These pages can be easily edited to suit the needs of students.

Topics include

Living things and ecosystems
Food and ecosystems
Plants in ecosystems

Animals in ecosystems
Plant and animal biodiversity
Humans and ecosystems

Associated titles in the *Achieve!* Instant Lessons series

- Science understanding
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 - Chemistry for years 9–10
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 - Biology for years 9–10
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 - Biology for years 7–8
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- Living things
- Indigenous contexts
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